Physics of Complex Systems Homework 9

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Problem 1 (The First Gap)**.** The Hamiltonian of the Ising quantum chain is defined by

$$H_N = \sum_{n=0}^{N-1} \left(\sigma_n^x \sigma_{n+1}^x + \lambda \sigma_n^z \right),$$

where $\lambda > 0$ and

$$\sigma_n^{x|y|z} = 1_{2^n} \otimes \sigma^{x|y|z} \otimes 1_{2^{N-n-1} \times 2^{N-n-1}}$$

are Pauli matrices acting at site n, assuming periodic boundary conditions. The first (or lowest) gap Δ_N is defined as the difference between the first two lowest-lying eigenvalues of H_N.

(a) In Markov processes, the first gap of $\mathcal{L}$ is the inverse leading relaxation time. In particle physics, the first gap measures the mass. What is the interpretation of the first gap of H_N in the present context?

(b) With the *Mathematica*® code fragment for the tensor product $\otimes$ given in the lecture notes,[1] write a code snippet to set up the Hamiltonian for arbitrary N and λ.

Cross-check: The eigenvalues for $N = 3$ (i.e. 8×8) are $\{-\lambda - 1,\ -\lambda - 1,\ \lambda - 1,\ \lambda - 1,\ -2\sqrt{\lambda^2 - \lambda + 1} + 1,\ 2\sqrt{\lambda^2 - \lambda + 1} + 1,\ -2\sqrt{\lambda^2 + \lambda + 1} + 1,\ 2\sqrt{\lambda^2 + \lambda + 1} + 1\}$.

(c) With (b), compute numerically the first gap of H_N for

$$N = 2, 4, 6, 8, 10, 12$$

and plot the gap for $\lambda = 0.8,\ 1,\ 1.2$ (double-logarithmically) as a function of N. For which λ do you get an almost straight line of points and why?[2]

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[1] You are of course free to use any other algebraic computer system as well.

[2] The execution for $N = 12$ should take less than 20 seconds. If not, go only up to $N = 10$.

(d) Compute the negative discrete logarithmic derivative

$$\nu(N) = -\frac{\ln[\Delta_{N+2}/\Delta_N]}{\ln[(N+2)/N]}$$

for $\lambda = 1$ and

$$N = 2, 4, 6, 8, 10,$$

and guess the value of the critical exponent ν, defined by

$$\nu = \lim_{N\to\infty} \nu(N).$$

Proof. (a) The first gap represents the excitation energy between the ground state and the first excited state.

(b) First we define the tensor product on Pauli matrices, using a **one-indexed convention** instead.

```
In[1]:= σ[k_, x_, n_] := KroneckerProduct[IdentityMatrix[2^(k - 1)], PauliMatrix[x],
         IdentityMatrix[2^(n - k)]]
```

Then, we define the Ising Hamiltonian, being careful to use a mod function to impose periodic boundary conditions

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In[2]:= IsingHamiltonian[n_] := Sum[σ[k, 1, n] . σ[Mod[k + 1, n, 1], 1, n] + λ σ[k, 3, n], {k, 1,
         n}]
```

We then define a function to find a spectral gap, given a list of eigenvalues

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In[3]:= spectralgap[lst_] := (Abs[#[[1]] - #[[2]]] &)[TakeSmallest[lst, 2]]
```

and finally we compute the spectral gaps as a function of n for the various λ.

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In[4]:= λ12 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 1.2], {n,
         2, 12,2}], n];

In[5]:= λ1 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 1.], {n, 2,
         12,2}], n];

In[6]:= λ08 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 0.8], {n
         , 2, 12,2}], n];
```

□

Problem 2 (DP Mean Field Approximation)**.** In the mean field approximation of directed percolation (DP) extended by a term for diffusion, the density of active sites $\rho(x,t)$ evolves according to the partial differential equation

$$\dot{\rho} = \lambda\rho(1-\rho) - \rho + D\nabla^2\rho.$$

(a) Determine the homogeneous (space-independent) stationary (time-independent) density

$$\rho_{\text{stat}}$$

as a function of λ. Where is the critical point λ_c? Test the solutions for stability by a stability analysis.

(b) Find the exponent for

$$\rho_{\text{stat}} \sim (\lambda - \lambda_c)^\beta$$

near the critical point.

(c) At criticality $\lambda = \lambda_c$, the homogeneous density $\rho(t)$ decays as

$$\rho(t) \sim t^{-\delta}.$$

Compute the exponent δ.

(d) Show that the Green's function of the mean field equation at the critical point can be written in the form

$$G(\mathbf{x},t) = t^{-1} f\left(\frac{x^2}{t}\right).$$

Hint: It is sufficient to derive an autonomous differential equation for f; you do not have to compute f explicitly. Restrict yourself to $t > 0$ and ignore the scaling of the δ-functions.

Proof. (a) If the density is homogeneous, the left hand side, as well as the second term on the right hand side vanishes and we are left with

$$\lambda\rho_{\text{stat}}(1-\rho_{\text{stat}}) - \rho_{\text{stat}} = 0.$$

We factor ρ out to find

$$0 = \rho_{\text{stat}}[\lambda(1 - \rho_{\text{stat}}) - 1]$$

which has either the solution $\rho_{\text{stat}} = 0$ or $\lambda(1 - \rho_{\text{stat}}) - 1 = 0$, which implies

$$\rho = 1 - \frac{1}{\lambda}.$$

$\square$